

Chamathsara Chamodie Marasingha

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Profile

Microsoft Certified Power BI Data Analyst Associate and Applied Data Science and Communication undergraduate with a strong foundation in data analysis, visualization, and business intelligence. Experienced in working with real-world datasets using Power BI, Excel, Python, and SQL to generate actionable insights. Passionate about combining data and communication to support informed decision-making, and eager to contribute to data-driven projects in an internship setting.

Education

General Sir John Kotelawala Defence University 2024 – 2027
Bachelor of Science in Applied Data Science and Communication

- **Relevant Coursework:** Advanced SQL and Cloud Database, Data Mining, Fundamentals of Mathematics and Statistics, Fundamentals of R, Effective Writing Skill, Python, Effective Presentation Skill, Economics

G.C.E. Advanced Level Examination – Biological Science Stream 2023
Subjects: Biology, Chemistry, Physics

Skills

Programming: SQL, Python, R
Data Analytics Visualization: Power BI, Excel: Power Query, Pivot Tables, Dynamic Functions (XLOOKUP, SUMIFS), Automated Dashboards, Pandas, NumPy, Seaborn, Matplotlib
Databases: Microsoft SQL Server (SSMS)
Soft Skills: Communication, Teamwork, Problem-Solving

Projects

Customer Demographics & RFM Analysis Dashboard | *Power BI, Excel, DAX, Data Modeling, Data Visualization* | 🌐 2026

- Developed a multi-page Power BI dashboard analyzing customer behavior through demographic insights and RFM (Recency, Frequency, Monetary) analysis across \$29.36M in total sales and 18.48K customers.
- Segmented customers into loyalty tiers (loyal, new, at-risk, and lost) using RFM scoring to support targeted retention campaigns and optimized marketing ROI.
- Visualized key metrics including gender distribution, marital status, age bands, and global sales by country to enable data-driven decision-making for customer engagement strategies.

AeroAI: Airline Delay Analysis Using Data Mining | *R, K-Means Clustering, ggplot2, factoextra* | 🌐 2025

- Analyzed U.S. airline flight delays for December 2019 and 2020 using Bureau of Transportation Statistics data to uncover key drivers of delay performance.
- Applied Min-Max normalization, Euclidean distance matrices, and K-Means clustering (Elbow Method, K=3) in R to segment airlines into high-, moderate-, and best-performing delay clusters.
- Visualized cluster characteristics using ggplot2, heatmap, and factoextra, identifying late aircraft and carrier-related issues as the leading causes of delays.
- Delivered data-driven recommendations on scheduling and crew turnaround to improve punctuality for airlines, airports, and policymakers.

Samsung Supply Chain & Logistics Dashboard | *Power BI, Excel, DAX, Data Modeling, Data Visualization* | 🌐 2026

- Built an interactive dashboard to analyze supplier performance, inventory turnover, shipment efficiency, and customer purchasing patterns across platforms.
- Identified procurement cost drivers, stock optimization opportunities, and delivery bottlenecks using DAX measures and data modeling.
- Visualized key KPIs including revenue & profit trends, supplier lead times, inventory stock value, and shipment costs to support data-driven supply chain decisions.

SDG 2 – Zero Hunger Analytics Dashboard | *Excel, SQL Server, Power BI, Advanced DAX* | 🌐 2025

- Transformed UN SDG datasets into a dashboard analyzing global food security, nutrition, and sustainable agriculture trends.
- Applied SQL, star-schema modeling, and DAX to identify disparities, vulnerabilities, and policy priorities for 2030 Zero Hunger targets.

Autonomous Supplier Intelligence Agent | *Python, LangChain, Groq LLaMA 3, Streamlit* | 🌐 2026

- Developed an autonomous AI agent for supply chain risk analysis, reducing supplier research time from 2–3 weeks to under 60 seconds.
- Built a LangChain ReAct agent with Tavily API and Groq LLaMA 3 to evaluate suppliers across reputation, financial, geopolitical, and delivery risks.
- Implemented real-time monitoring with APScheduler and automated alerts, along with PDF report generation via ReportLab and Gmail SMTP.
- Designed an interactive Streamlit dashboard with supplier watchlist, persistent storage, and downloadable reports.

AI Policy Summarizer & Scenario Generator | *Python, Gradio, Gemini API, NLTK, HuggingFace* | 2026

- Built a dual-panel web application that extracts, preprocesses, and summarises real-world government policy documents using NLTK tokenisation, stopword removal, and lemmatisation.
- Integrated Gemini 2.5 Flash API and a BART transformer fallback to generate structured, scenario-adapted policy drafts across six predefined lenses including Education, Healthcare, and Rural Development.
- Implemented a custom scenario engine allowing users to define arbitrary adaptation contexts, producing formally structured six-section policy outputs derived exclusively from uploaded source documents.

Certifications

Microsoft Certified: Power BI Data Analyst Associate (PL-300) | Microsoft

Demonstrated expertise in data analysis, data modeling, visualization, and creating actionable insights using Power BI.

Microsoft Certified: Azure Data Fundamentals (DP-900) | Microsoft

Gained foundational knowledge of core data concepts, cloud data services, and Azure data solutions.

Python for Data Science, AI & Development | IBM (Coursera)

Developed strong Python programming skills and applications in data science and AI.

Certificate in Quantitative Data Analysis | University of Sri Jayewardenepura

Strengthened understanding of statistical and quantitative data analysis techniques for real-world applications.

Prompt Engineering for ChatGPT | Vanderbilt University (Coursera)

Learned to design effective prompts for AI tools, enhancing practical AI application skills.

Clubs & Extracurricular Activities

AI and Data Science Club — Co-Editor (2025–2026)

Contributed to content creation, organized workshops and seminars, and supported club activities for knowledge sharing in AI and data science.

References

Ms. Chamodhi Wickramasinghe

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